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Effect of plasma parameters on the properties of V_2O_5 deposited by plasma-enhanced atomic layer deposition

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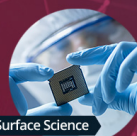
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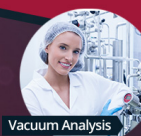
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Effect of plasma parameters on the properties of V_2O_5 deposited by plasma-enhanced atomic layer deposition

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ABSTRACT

In this work, we investigate the deposition of VO_x thin films by plasma-enhanced atomic layer deposition. Specifically, we focus on the effects of the plasma operating conditions, including applied power, O_2/Ar gas flow ratio, pressure, and coupling mode on the resulting thin film crystallinity and stoichiometry. In addition, we characterize the plasma and correlate the findings to the changes observed in the produced VO_x thin films. The majority of conditions investigated yield crystalline V_2O_5 . However, by increasing the pressure during the plasma half cycle, the V_2O_5 film can be deposited with a reduction in the crystalline character without any significant changes in stoichiometry. Langmuir probe measurements reveal that both the ion flux and plasma potential decrease with increasing pressure during the plasma step, which results in a corresponding drop in the energy flux density and ion energy delivered to the surface of the thin film. Analysis of the various plasma conditions investigated shows that increasing the pressure uniquely resulted in a decrease in the energy flux density and ion energy below a critical barrier necessary for complete crystallization of the deposited VO_x films.

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I. INTRODUCTION

Atomic layer deposition (ALD) of metal oxides is a cyclic chemical vapor deposition technique in which a gaseous metal-organic precursor and oxygen-containing reactant gas are sequentially dosed into a reactor to deposit a thin film in a layer-by-layer manner. Due to the self-limiting nature of each precursor and reactant exposure, ALD allows for highly uniform films to be deposited, even on high aspect ratio surfaces. Plasma-enhanced ALD (PEALD) utilizes a plasma during the reactant half-step, offering numerous benefits including enhanced crystallinity, especially at low substrate temperatures, and access to metastable phases.

As discussed in detail by Knoops *et al.*¹ and Boris *et al.*² ALD chambers and plasma sources come in a diverse range of geometries with a wide processing space in which gas flow, gas chemistry, applied power, chamber pressure, and coupling mode all impact the type of reactive species and amount of energy that will be delivered

to the substrate/thin film surface during deposition. The importance of these parameters has been highlighted in numerous recent works. For instance, Wheeler *et al.* demonstrated that the crystallinity and phase of Ga_2O_3 were selectively controlled through the alteration of the plasma parameters during deposition. Characteristics of the deposited material were correlated with specific characteristics of the plasma, such as plasma potential and ion flux.³ Similarly, the rutile and anatase phases of TiO_2 have been selectively deposited by PEALD by altering the plasma chemistry (pure O_2 versus O_2/Ar mixture).⁴ Zhang *et al.* observed that the growth-per-cycle (GPC), crystallinity, and mobility of In_2O_3 thin films all improved with increasing plasma power; these results were tied to an increase in the O^+ concentration and plasma potential at higher plasma powers.⁵ Werbrouck *et al.* carried out diagnostics of O_2 plasmas and studied the resultant effects of the plasma on deposited Al_2O_3 films.⁶ Their findings, which are relevant to this work, include a linear dependence in the amount of produced plasma species with plasma power

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and a decay in ion flux delivered to the surface with increasing pressure. These works motivate further investigation into establishing the critical parameters of reactive plasmas that can be used to control the properties of deposited thin films, all of which will be material- and application-specific.

The V-O system is a good materials system to investigate the impact of plasma parameters on crystallinity and stoichiometry due to the complexity of its phase diagram, where over 30 competing compounds of various stoichiometries are known to exist.⁷ Furthermore, numerous crystalline compounds, such as VO₂ and V₂O₃, undergo metal-to-insulator transitions (MITs) as a function of temperature. Crystalline VO₂ is of particular interest due to its MIT occurring near room temperature ($\approx 68^\circ\text{C}$), making it an interesting candidate for many passive and active functional applications including energy efficient windows,^{8,9} transistors,^{8,9} bolometers,⁹ and neuromorphic devices.¹⁰ A VO₂ thin film traditionally produced by thermal ALD has its stoichiometry controlled by the oxidation state of the metal-organic precursor. Therefore, a V⁴⁺ precursor, such as tetrakis(ethylmethylamino)vanadium (TEMAV), is typically used in the process. However, VO₂ films produced by thermal ALD are amorphous in the as-deposited state and require postdeposition annealing steps at relatively high temperatures to crystallize, where oxygen partial pressure, temperature, and time are critical to maintaining the VO₂ stoichiometry established by the thermal ALD process.¹¹

Assuming that sufficient control of the plasma properties can be achieved, PEALD offers an opportunity to produce crystalline VO₂ in a single-step process at low temperature. However, previous attempts were unsuccessful due to the aggressive oxidizing characteristics of O₂ plasmas, resulting in the formation of V₂O₅ regardless of the oxidation state of the metal-organic precursor used.¹² Recently, Mohammad *et al.* have investigated the effects of varying plasma conditions on the deposition of VO_x using O₂ plasmas, specifically studying the effect of plasma power (100–300 W) and plasma chemistry (10 SCCM O₂, 50 SCCM O₂, and 50 SCCM O₂/20 SCCM Ar).¹³ The authors also investigated the impact of a capacitively coupled versus inductively coupled plasma (ICP) on the stoichiometry and crystallinity of the films deposited. Across numerous deposition conditions tested, the authors found that V₂O₅ was formed, and postdeposition annealing steps were required to reduce V₂O₅ into substoichiometric phases. While the authors identified differences in the GPC and refractive index of V₂O₅ by changing the type of plasma source, absent from this study are measured properties of the plasma source itself and how those changes in the plasma correlate with changes in the material.

In this work, we investigate the effect of plasma parameters on the deposition of VO_x by PEALD. We focus on how the various plasma parameters affect the structure and stoichiometry of films produced under a broader range of conditions than those explored by Mohammad *et al.*¹³ Additionally, we apply a suite of plasma diagnostics to establish correlations between plasma characteristics and properties of the VO_x thin films. While most plasma conditions are found to produce crystalline V₂O₅, by altering the pressure during the plasma half-step, we observe that the degree of crystallinity decreases with increasing pressure for inductively coupled plasmas. Through the analysis of the ion flux and plasma potential under the various plasma conditions, it is determined that increasing the pressure uniquely resulted in a decrease in the

energy flux density and ion energy below a critical barrier necessary for complete crystallization of the deposited VO_x films.

II. METHODS

A. Material deposition

VO_x films were deposited on double-side polished c-Al₂O₃ and Si witness substrates at 150 °C for 200 cycles by PEALD in a Kurt J. Lesker Company ALD150LX system. Tetrakis(ethylmethylamino)vanadium(IV) (TEMAV) (STREM Chemicals, 98%) was used as the vanadium precursor (105–110 °C, 60 ms pulse time, 10 s purge time), and an O₂/Ar plasma was used as the oxidizing reactant source. For this study, O₂/Ar plasma parameters were varied, including pulse duration (5–80 s), power (50–950 W), pressure during the plasma step (850–5000 mTorr), and O₂/Ar flow ratio (10:160, 50:120, and 90:80 SCCM). The plasma purge step was 5 s in all cases. The base pressure of the system under typical ALD gas flows is ≈ 850 mTorr, and variations from this pressure were controlled with a throttle valve located downstream of the process chamber (upstream of the processing pump).

B. Material characterization

Film deposition is monitored in real-time using an *in situ* spectroscopic ellipsometer (Woollam M-2000) and postdeposition using an *ex situ* variable angle spectroscopic ellipsometer (Woollam alpha-SE). Film thicknesses are extracted through the analysis of the transparent region of the spectra (500 nm and above) and fit with a Cauchy model. Stoichiometry and impurity concentrations were determined with a Thermo Scientific NEXSA x-ray photoelectron spectroscopy (XPS) system equipped with an Al K α x-ray source (1486.7 eV) and Ar flood gun for charge neutralization during data acquisition. Depth profiles were obtained on all samples using an Ar cluster source (500 atom cluster size, 6000 eV cluster energy) for 900 s. The Ar cluster source size and energy were optimized to remove surface contamination while limiting preferential sputtering to reveal the nature of the bulk of the film deposited. Raman spectroscopy was carried out using a Thermo Scientific DXRxi microscope with a 532 nm laser for the nondestructive qualification of the film phase. Laser power was maintained at 0.5 mW, and acquisition times were 60 s.

C. Plasma characterization

Charged particle characteristics near the substrate holder were determined using a Langmuir probe from Impedans Ltd. (Fig. 1). The probe is radio frequency (RF) compensated with self-resonant inductive filters at 0.4, 2.0, 13.56, 27, and 60 MHz to eliminate the effects of plasma potential fluctuations due to RF coupling. The ion density is calculated using the models of ion current collection published by Allen *et al.*¹⁴ and Chen¹⁵ and accounts for the effects of ion-neutral collisions within the sheath. This is relevant for the pressure range being explored in this work. The plasma potential was determined using the zero-crossing of the second derivative of the I-V trace taken by the probe.

Optical emission spectroscopy (OES) of the ICP source was performed using two Ocean Insight spectrometers, HR4 and HR2000+, which captured spectra in the range of 650–950 nm.

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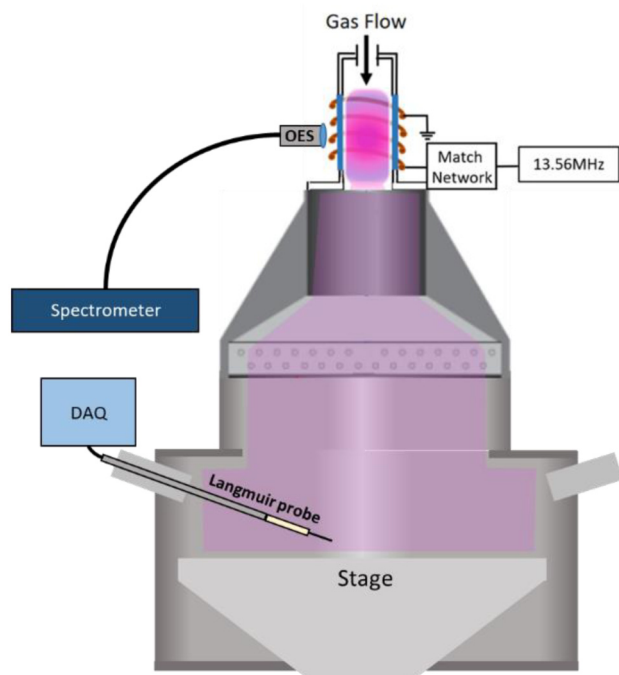


FIG. 1. Diagram of the remote inductively coupled plasma source in the PEALD system used in this work. The figure highlights the different locations characterized by the OES (remote plasma source) and Langmuir probe (downstream of the remote plasma near sample stage).

From each spectrum, the production of atomic oxygen was estimated through actinometry. The atomic oxygen density was determined by measuring the ratio of emission intensities between the 844 nm line, corresponding to the $O(3p^3P)$ state of atomic oxygen, and the 750 nm line, associated with the $2p_1$ level of argon. This ratio was used to approximate the atomic oxygen density based on the procedure outlined by Saloum *et al.*¹⁶ The rate constants were obtained from Fuller *et al.* by assuming that the electron temperature was 1.5 eV.¹⁷

III. RESULTS AND DISCUSSION

A. Effect of plasma pulse duration

The plasma pulse duration was varied from 5 to 80 s for two different pressure conditions during the plasma exposure step (850 and 1600 mTorr), and the resulting GPC extracted from both *in situ* and *ex situ* spectroscopic ellipsometry is presented in Fig. 2(a). The two pressure conditions showed similar behavior, albeit at different time scales. Both exhibit an initial rapid decrease in GPC with increasing plasma pulse duration followed by a more gradual decay before saturating at ≈ 0.055 nm/cy. Saturation at 1600 mTorr (60 s) took twice as long as at 850 mTorr (30 s).

Raman spectra of the $VO_x/c\text{-Al}_2O_3$ thin films produced at a pressure of 850 and 1600 mTorr are presented in Figs. 2(b) and 2(c), respectively. Again, similar behavior between the two pressure

conditions is observed at differing time scales. The Raman spectra saturate at a plasma pulse duration of 9 s at 850 mTorr and 15 s at 1600 mTorr; under both pressure conditions, the saturated Raman spectra are consistent with V_2O_5 (α -phase).¹⁸ At a 5 s plasma pulse duration, both pressure conditions resulted in Raman spectra dominated by features associated with the $c\text{-Al}_2O_3$ substrate and some weak features consistent with the formation of V_2O_5 , including the strongest peak at $\approx 147\text{ cm}^{-1}$ and broad peaks at approximately 300 and 475 cm^{-1} . At 1600 mTorr, the Raman spectra after a 9 s plasma pulse duration show a mix of substrate features and broad features consistent with V_2O_5 , indicating a mix of crystalline and amorphous V_2O_5 . XPS analysis indicates that the C and N contamination in the sample (from residual incorporated TEMAV precursor ligands and/or redeposited of dissociated reaction by-products¹⁹) dropped significantly with increasing plasma pulse duration [Fig. 2(d)]. At 850 mTorr, the contamination leveled off at 9 s with C contamination of <2 atomic % and N of about 0.5 atomic %. At 1600 mTorr, the contamination decreased to similar levels as 850 mTorr at a plasma pulse duration of 20 s.

The results obtained from studying the effect of the plasma exposure time suggest that a critical plasma pulse duration is necessary to complete ligand exchange and establish an ALD window. Operation below this critical exposure time threshold results in a higher GPC, reduced levels or elimination of crystallinity, and higher C and N contamination. A similar decrease in the GPC with increasing plasma exposure time has been demonstrated by Haider *et al.* and was explained as a decrease in contamination from precursor ligands.²⁰ This explanation is consistent with the results obtained in our work, which shows a decrease in both GPC and contamination that eventually saturate with greater plasma exposure times. Above the critical exposure threshold, films produced are crystalline V_2O_5 with a saturated GPC of 0.055 nm/cy, and minimal C and N contamination (<2 at. %). Additionally, the critical plasma pulse duration necessary for saturation exhibits a direct relationship with pressure. The results here suggest that the time to saturate is proportional to pressure, as roughly a twofold increase in pressure required twice the time to reach saturation.

B. Effect of plasma power

The effect of plasma power was investigated over 50–950 W. For this study, the plasma pulse duration (9 s), O_2/Ar flow through the ICP (10 SCCM O_2 , 160 SCCM Ar), and system pressure (850 mTorr) were kept constant. The structural and chemical makeup of the thin films was investigated with Raman spectroscopy and XPS, which are presented in Figs. 3(a) and 3(b), respectively. Similar Raman and XPS spectra are observed throughout the entire power range investigated. The Raman spectra show the features characteristic of V_2O_5 with some additional substrate features.

In situ plasma characterization was carried out to further understand the effects of plasma power. Figure 4(a) shows the atomic O density in the plasma measured by OES as a function of power, while Fig. 4(b) shows the ion flux and the plasma potential measured with a Langmuir probe under similar conditions. The coupling nature of the plasma was found to vary with plasma power. Low power plasmas were capacitively coupled (denoted by gray region in each plot), while higher power resulted in an inductively

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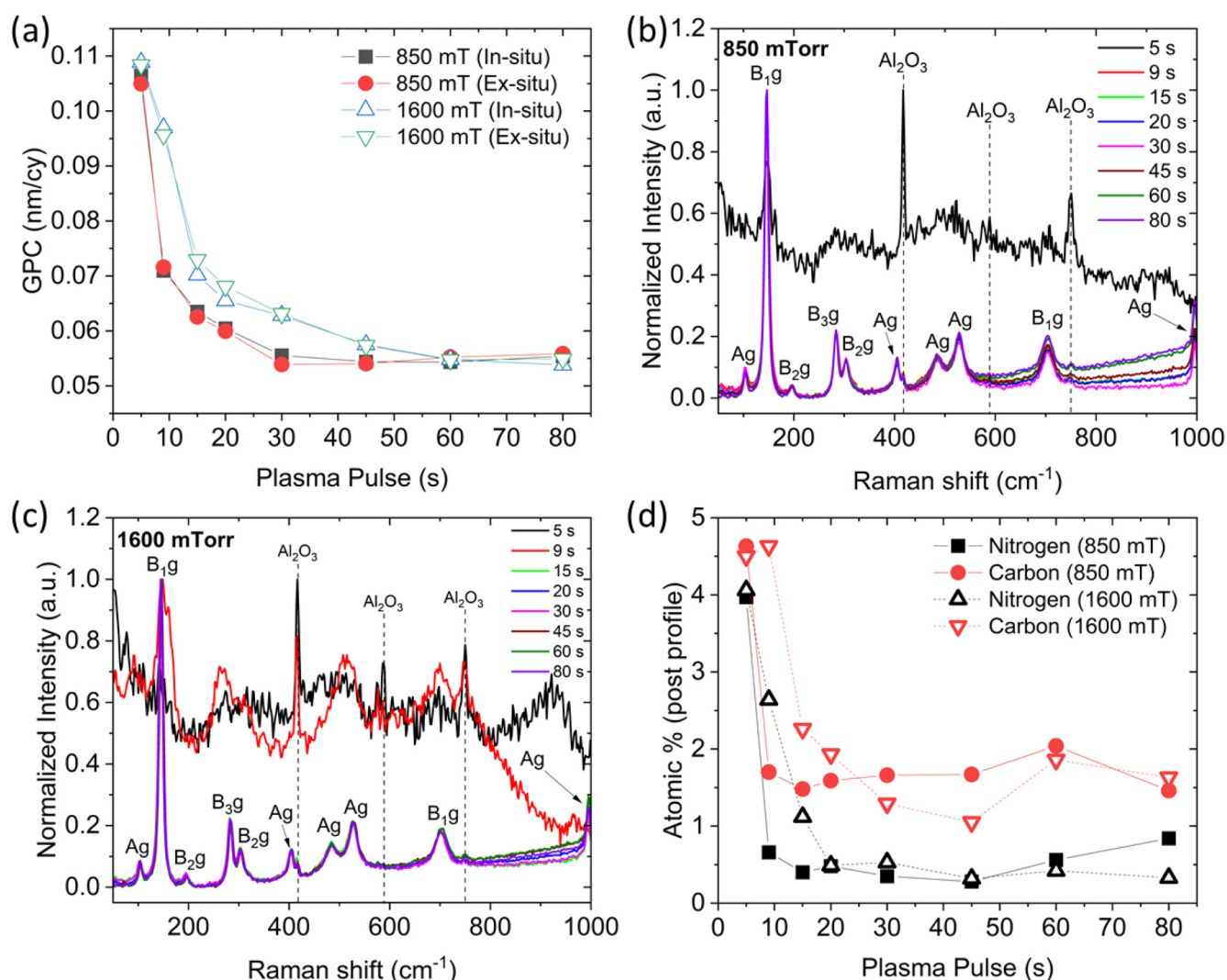


FIG. 2. (a) GPC of VO_x as a function of plasma pulse duration at 850 and 1600 mTorr. Normalized Raman spectra of VO_x deposited as a function of plasma pulse duration at 850 (b) and 1600 mTorr (c). (d) XPS-derived N and C atomic % as a function of plasma pulse duration at 850 and 1600 mTorr.

coupled plasma. It is worth noting that the OES and Langmuir data presented here were collected months apart. Over this time, it was found that the plasma power threshold separating a capacitively coupled and inductively coupled plasma shifted (likely due to system drift/aging). The gray regions denoting this behavior differ in each plot and are representative of the system behavior on the day each respective dataset was collected. In the range of 150–950 W, the atomic O density increased by two orders of magnitude, ranging from 3.2×10^{18} to 1.7×10^{20} atoms/m³. While the ion flux was lowest at 15 W (6.6×10^{15} m²/s) and highest at 1000 W (1.0×10^{18} m²/s), a local maximum in the ion flux was observed at 120 W (below the capacitively/inductively coupled transition). The plasma potential was minimum at 150 W (13.0 V) and maximum at 120 W (57.8 V).

In the range of 150–800 W, it was observed that the plasma potential increased with increasing power and plateaued in the 800–1000 W range.

The results obtained throughout the range of plasma powers explored here indicate that in all cases, crystalline V_2O_5 is produced. Furthermore, the overlapping XPS spectra indicate that no detectable changes in material stoichiometry occur. It can be concluded that the minimum atomic O density measured here (3.2×10^{18} atoms/m³) is sufficient to produce a fully saturated oxide (i.e., V^{5+}). Therefore, the addition of more O at higher plasma powers has little effect on the chemical nature of the deposited material. Additionally, the chemistry and structure were consistent in both inductively coupled and capacitively coupled plasmas; this

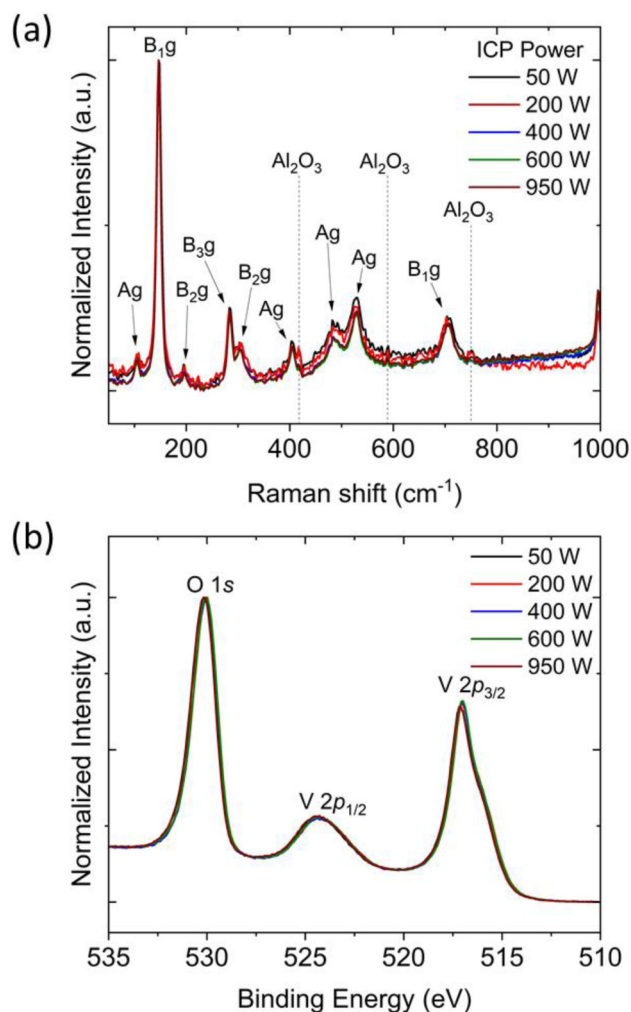


FIG. 3. (a) Normalized Raman spectra and (b) XPS V 2p and O 1s peaks of VO_x deposited at 850 mTorr with varying plasma power.

is consistent with the work of Mohammad *et al.*, in which they observed the formation of V_2O_5 using both inductively coupled and capacitively coupled hollow cathode plasma sources.¹³ In comparison with the work reported by Werbrouck *et al.*, which was limited to the 50–250 W range, we explore a significantly wider range in plasma power. Despite this, our results show a similar trend to that previously reported. The atomic O density as a function of plasma power reported here shows a trend that is roughly linear, albeit with some uncertainty due to the limited number of data points. This suggests that the trend observed by Werbrouck *et al.* extends beyond the low power regime investigated in their work. Despite the lack of changes in the stoichiometry and crystallinity of the produced material, it is worth noting how the properties of the plasma, specifically the ion flux and plasma potential, change across the capacitively coupled to inductively coupled mode

transition. Both the ion flux and plasma potential values were found to exhibit local maxima in the capacitively coupled mode just below the coupling mode transition.

C. Effect of plasma chemistry

The O_2/Ar ratio through the ICP during the plasma step was varied while keeping the same total flow (170 SCCM). For this study, the plasma pulse duration (9 s), plasma power (400 W), pressure (850 mTorr), and coupling mode (inductively coupled) were kept constant. The Raman spectra of the thin films as a function of gas ratio in the ICP are presented in Fig. 5(a). The three Raman spectra overlap nearly perfectly, suggesting that the films are structurally indistinguishable and all match the expected spectra of crystalline V_2O_5 . The GPC showed a subtle decrease with increasing O_2 flow fraction [Fig. 5(b)].

The OES-derived atomic O density as a function of O_2 flow fraction is presented in Fig. 6(a) and shows only subtle changes with changing gas chemistry and no consistent trend. Interestingly, this result implies that the percentage of dissociated O_2 (into atomic O) shows a large dependence on flow fraction with the highest percentage occurring at low O_2 flow fractions. The ion flux also shows no trend with varying O_2 flow fraction [Fig. 6(b)]. However, the plasma potential increased linearly with increasing O_2 flow fraction ranging from about 20–50 V over the flow ranges tested here [Fig. 6(b)].

Out of the flow ratios investigated here, the density of atomic O was at a minimum with a 0.06 O_2 flow fraction (2.39×10^{19} atoms/ m^3). Given that this value is still greater than the minimum atomic O density necessary to form V_2O_5 (as determined in Sec. III B), it reasons that under all O_2/Ar ratios tested here, V_2O_5 will be deposited, which the Raman results reported in this section confirm. Additionally, in this processing space, the ion flux to the sample surface was relatively constant, which in part explains the lack of significant variation in the film structure. The increasing plasma potential at higher O_2 flow fractions increases the average ion energy impacting the surface. However, the minimum value here (≈ 20 V) is sufficient to deliver ions capable of crystallizing the V_2O_5 film. Therefore, increasing the plasma potential further does not have a noticeable effect on the structure of the film. However, the decrease in the GPC with increasing O_2 flow fraction shows a correlation with the increase in plasma potential. It is plausible to assume that this is due to increased ion energies delivered to the sample surface, which may result in the liberation of surface species through ion-driven etching. However, it should be noted that the decrease in the GPC with increasing O_2 flow fraction is relatively small.

D. Effect of pressure

The impact of operating pressure was investigated by varying the pressure from 850 to 5000 mTorr while keeping the plasma pulse duration (15 s), power (400 W), and O_2/Ar flow through the ICP (10 SCCM O_2 , 160 SCCM Ar) constant. The throttle valve downstream of the processing chamber was utilized to increase the pressure during the plasma step. Raman spectra for VO_x films produced with inductively coupled plasmas under various pressures are presented in Fig. 7(a). The data show a decrease in the intensity

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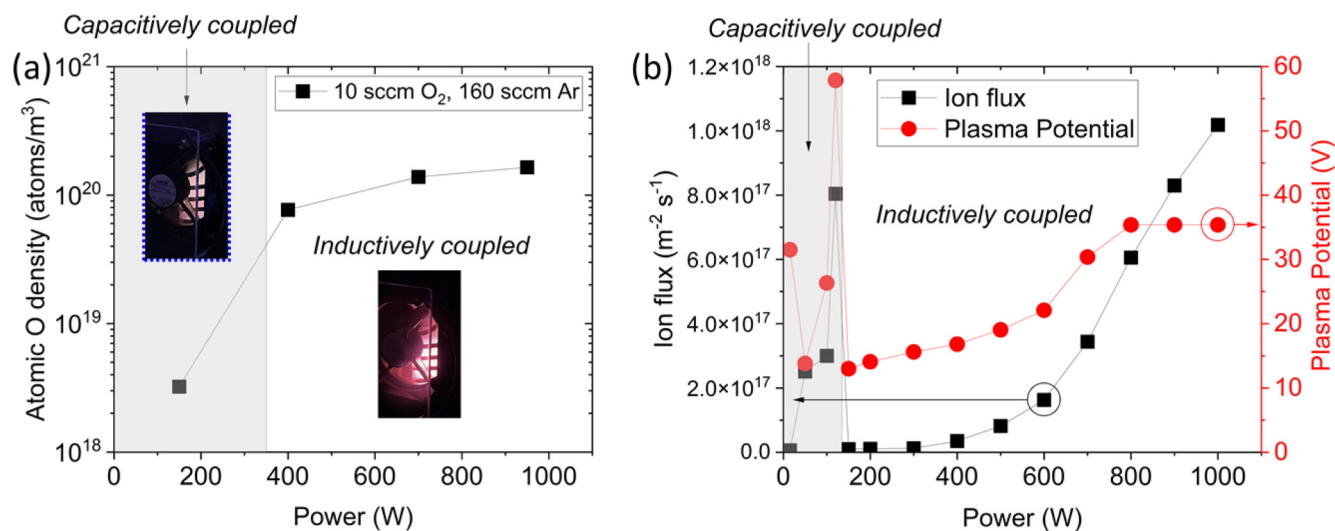


FIG. 4. (a) Atomic O density vs plasma power measured with OES. (b) Ion flux and plasma potential as a function of plasma power. Gray regions of each plot represent processing space in which the plasma operated in a capacitively coupled mode.

of Raman features associated with V₂O₅ as pressure is increased while the intensity of the substrate peaks remains unchanged. At 5000 mTorr, the plasma could be selectively operated in either an inductively coupled or capacitively coupled mode. Using an inductively coupled plasma, the Raman features associated with V₂O₅ are below the limit of detection [Fig. 7(b)]. In contrast, the use of a capacitively coupled plasma results in strong Raman signatures of V₂O₅. XPS results as a function of pressure are shown in Figs. 7(c) and 7(d). Within the pressure range tested, most films have similar O 1s and V 2p core level spectral line shapes [Fig. 7(c)] and C and N contamination [Fig. 7(d)]; the exception is the film deposited at 5000 mTorr while operating in the inductively coupled mode, which resulted in an increased width of the O 1s and V 2p as well as a significant increase in the degree of C and N contamination.

Figure 8(a) shows the OES-derived atomic O concentration in the plasma at various pressures. Generally, it is observed that the atomic O concentration increases with increasing pressure, and a density of 2.1×10^{19} – 3.5×10^{20} atoms/m³ is observed for the pressure range investigated. Furthermore, it was observed that at the highest pressures, the atomic O concentration produced by inductively coupled plasmas was modestly greater than that produced by capacitively coupled plasmas. Figure 8(b) shows the ion flux near the sample surface and plasma potential measured using a Langmuir probe over the same pressure range. The data show a decrease in ion flux from 4.2×10^{16} to 9.7×10^{15} ions/m²s ($\approx 77\%$ decrease) over the measured range for inductively coupled plasmas. Notably, the most significant decrease in ion flux is measured between 850 and 1600 mTorr, and only a minor decay is observed in the ion flux in the range of 1600–5000 mTorr. Over the same pressure range, the plasma potential increases slightly between 850 and 1600 mTorr and then decreases markedly from as high as from 22 V (1600 mTorr) to 9 V (5000 mTorr). At 5000 mTorr, it was

observed that both the ion flux and plasma potential increased substantially when operated in a capacitively coupled mode compared to an inductively coupled mode. This result is consistent with the behavior observed across the coupling mode transition discussed earlier in this paper; i.e., capacitively coupled plasmas result in a greater ion flux and plasma potential under otherwise similar conditions (power, flow rates, etc.) [see Fig. 4(b)].

In the context of inductively coupled plasmas investigated here, increasing the pressure during the plasma half-step resulted in a decrease in the degree of crystalline V₂O₅ in the sample, a decrease in the ion flux delivered to the sample, and a decrease in the plasma potential. Ions are known to carry a significant amount of the energy to the sample, and thus, a decrease in ion flux results in a decrease in the energy delivered to the sample surface.² Furthermore, the kinetic energy of those ions is limited by the plasma potential, and so a decrease in plasma potential results in a reduction in kinetic energy. The combined effect is that the energy flux density delivered by ions to the sample decreases with increasing pressure of the system. Thus, the decrease in crystallinity observed at 2500 and 3700 mTorr could be due to this reduced energy flux density being below a crystallization energy barrier, since all samples deposited from 850 to 3700 mTorr exhibit a similar chemical composition and bonding environment [Figs. 7(c) and 7(d)]. It is worth noting that the lack of crystallinity observed at the highest pressure investigated (5000 mTorr) is accompanied by significantly higher levels of C and N contamination in the sample and a distinctly different O and V chemistry relative to the films produced at lower pressures. The increased contamination levels found associated with this deposition condition can be expected to provide an additional barrier to material crystallization. These changes in chemistry suggest either incomplete ligand exchange or redeposition of dissociated reaction by-products.¹⁹ The

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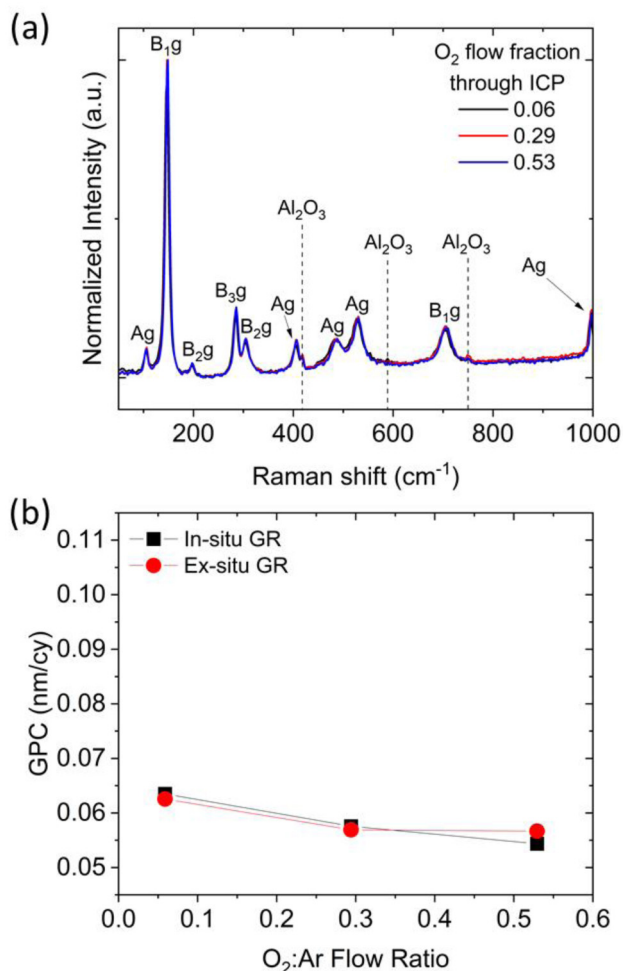


FIG. 5. (a) Raman spectra of VO_x deposited with varying O₂ flow fraction. (b) GPC as a function of O₂ flow fraction. The y axis plotted on the same scale as Fig. 2(a) to highlight the small magnitude of change across the O₂ flow fraction range investigated.

former may be attributed to a decrease in the energy flux density delivered to the sample surface or a reduced flux of reactive oxygen species and the latter due to an enhanced residence time associated with the high-pressure environment.¹⁹ While the density of atomic oxygen in the plasma source increased with pressure [Fig. 8(a)], it is reasonable to expect that the downstream flux of atomic oxygen available at the surface will decrease with increasing pressure due to the changes in diffusion and density gradients. As such, a reduced energy flux density, reduced atomic oxygen flux, and enhanced residence time could each contribute to the absence of crystallinity in this sample. However, the observation of crystalline V₂O₅ with a low impurity content deposited with a capacitively coupled plasma at 5000 mTorr suggests that there is, in fact, both sufficient atomic O delivered to the sample surface necessary for the chemical

saturation of V₂O₅ [Fig. 8(a)] and a sufficiently low residence time at this pressure for either coupling mode. From this, it is concluded that a significantly lower energy flux density in the case of the inductively coupled plasma is the primary cause for the absence of crystallinity and incomplete ligand exchange in the film deposited at 5000 mTorr. The additional energy provided by the capacitively coupled plasma likely aids in producing films with less C and N impurities than the film deposited with an inductively coupled plasma.

E. Effect of ion energy and flux on crystallinity

Of the variables investigated in this work, the pressure during the plasma half-step and the coupling mode of the plasma had the greatest impacts on the structural properties of the deposited thin films. Specifically, an increase in pressure results in a decrease in the crystallinity of the deposited films. However, highly crystalline films could be produced at 5000 mTorr when the plasma is operated in a capacitively coupled mode. Changes in neither power or relative gas flows, within the ranges explored in this work, resulted in such structural differences. Plasma characteristics measured by a Langmuir probe (ion flux and plasma potential) and OES (atomic O density) across changes in pressure, power, and relative gas flow showed comparable variations, thus complicating the understanding of why significant changes in crystallinity were not observed for all variables. To better understand this, we consider the work of Boris *et al.*, which argues that the energy flux density delivered to the surface of a growing film can be used to drive crystallinity.² Figure 9 follows their description, where the energy flux density delivered by ions impacting the surface is the product of the flux of ions and their kinetic energy. Here, however, we have substituted the plasma potential for the ion energy, which, in the absence of measured ion energies, serves as a reasonable proxy. The upper right section of this plot is the processing space of highest energy flux density, and the lower left section represents the region of lowest energy flux density. Lines of constant energy flux density extend from the upper left to the lower right of the plot and are represented as black dotted lines. In this plot, the measured ion flux and plasma potential at different deposition pressures in the present study are overlaid. The data point furthest to the lower left (5000 mTorr) exhibits the lowest energy flux density and corresponds to the deposited film that showed no signs of crystalline character. Alternatively, points further to the upper right (≤ 3500 mTorr) correspond to conditions that produced evidence of crystalline V₂O₅. The data point corresponding to the crystalline film produced with a capacitively couple plasma at 5000 mTorr is also included and represents the highest energy flux density. The combined results suggest the presence of a critical energy flux density for the onset of crystallization, which is approximated by the red dashed line in the plot.

While Fig. 9 provides a qualitative understanding of the results, it does not properly consider the energy delivered to the surface via the flux of *all* species. Indeed, at these pressures, the sheath is considered collisional. That is, not all ions leaving the bulk plasma will arrive at the surface with an energy commensurate to that of the plasma potential, V_p . The fraction of ions that arrive at the surface

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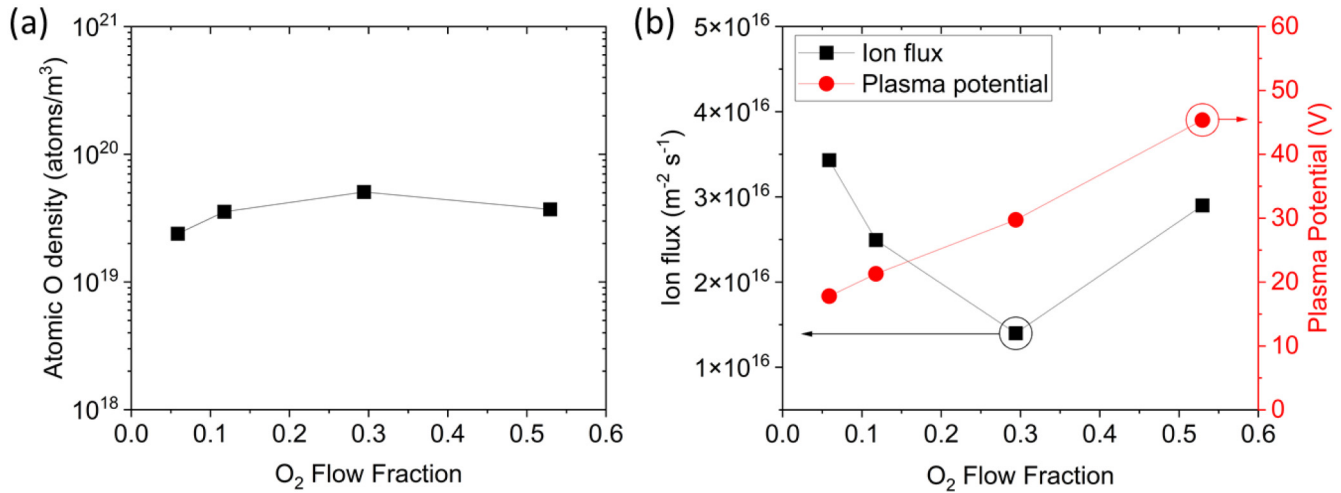


FIG. 6. (a) Atomic O density vs O₂ flow fraction measured with OES. The y axis plotted on the same scale as other OES measurements for comparison. (b) Ion flux and plasma potential as a function of O₂ flow fraction.

without a collision, I/I_o , can be expressed as

$$\frac{I}{I_o} = e^{-n_{\text{gas}}\sigma_{\text{ion}}s}, \quad (1)$$

where n_{gas} is the neutral gas density, σ_{ion} is the scattering cross section, and s is the sheath thickness. It is estimated that slightly less than 50% of ions transit the sheath without collisions at 850 mTorr; that value falls to less than 2% above 2400 mTorr. The average ion energy (E_{ion}) at the surface due to collisions in the sheath can be estimated following the work of Lieberman and Lichtenberg,²¹ using

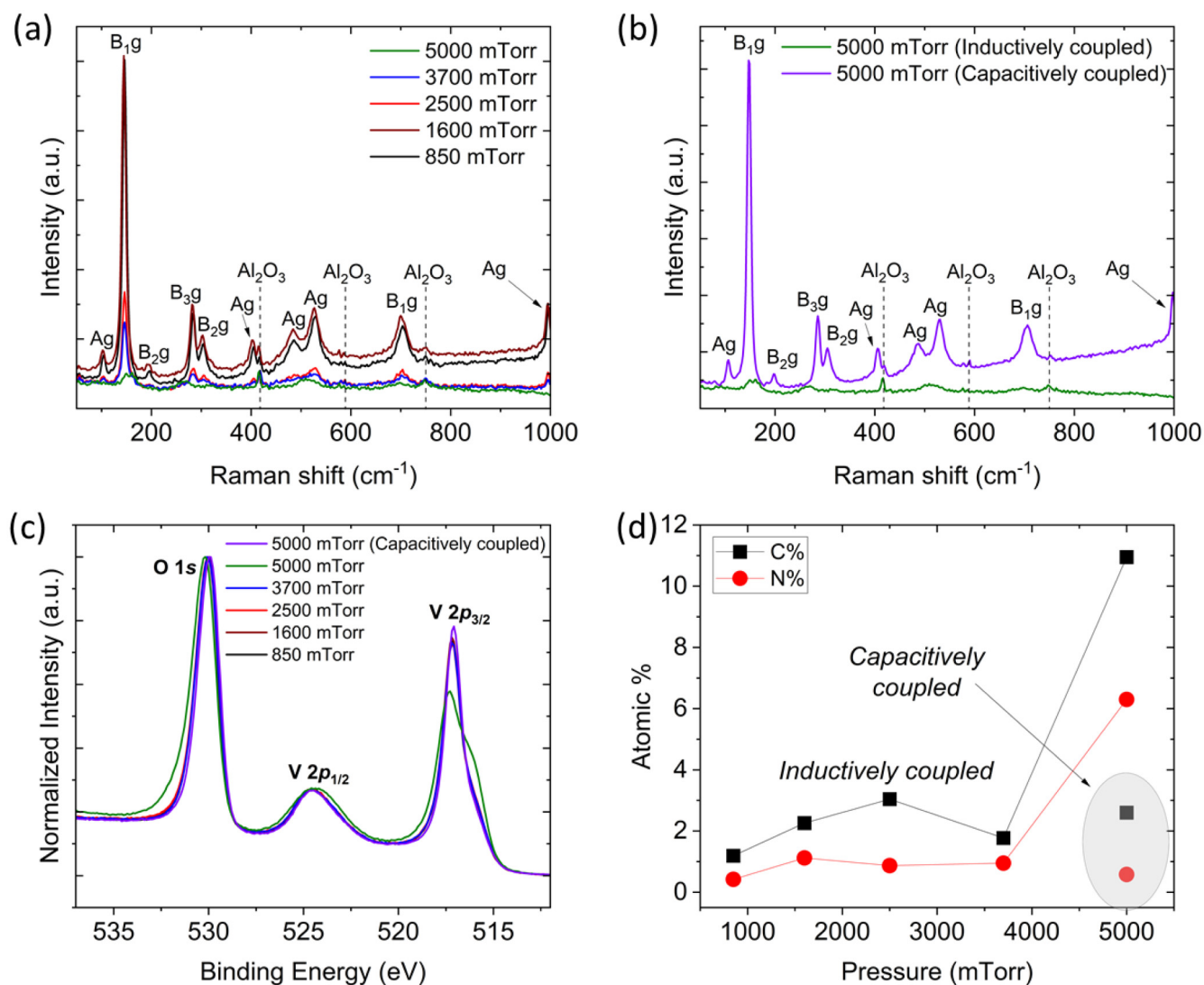
$$E_{\text{ion}} \approx 0.62 \left(\frac{\lambda_{\text{ion}}}{s} \right) V_p, \quad (2)$$

where $\lambda_{\text{ion}} = 1/n_{\text{gas}} \sigma_{\text{ion}}$. Because the mean free path decreases with increasing pressure, it is clear that using the plasma potential in Fig. 9 fails to capture the full change in energy flux density as pressure varies. However, it is important to recognize that during such collisions, a fast neutral (FN) is produced, which can also deliver energy to the surface. As such, the energy flux density (and charge) delivered to the surface by ions and fast neutrals in a collisional sheath will be the product of the ion flux and V_p . This is illustrated in Fig. 10 for the case of an argon ion undergoing a single charge exchange collision (e.g., $\text{Ar}^+ + \text{Ar} \rightarrow \text{Ar}_{\text{FN}} + \text{Ar}^+$) occurring at a position (x) in the sheath with a potential of $\frac{1}{2}V_p$. Each particle will arrive at the surface with a kinetic energy of $\frac{1}{2}V_p$. It is akin to moving along a line of constant energy flux density from low to high ion flux in Fig. 9. An advantage in processing is that while the total energy delivered remains the same, the kinetic energy of each particle is reduced, thus potentially avoiding or at least minimizing particle-driven etching (e.g., sputtering) or damage (e.g., dislocations).

Ultimately, the energy flux density delivered by only ions [using Eq. (2) to determine E_{ion}] and the combined energy flux density of ions and fast neutrals (using the product of the ion flux and V_p) can be calculated from the various measurements in this work involving the pressure, power, and O₂ flow fraction of the plasma, as shown in Fig. 11. The largest variation in the energy flux density was found for the power series measurement (nearly 3 orders of magnitude in variation), while the smallest variation was found for the O₂ flow fraction dataset (less than 1 order of magnitude variation). Importantly, the variation in the pressure series provides the lowest energy flux densities. Of course, these simple calculations do not capture the complexities associated with high-pressure oxygen mixtures, where inelastic collisions and scattering processes in the sheath reduce the energy delivered to the surface. Accordingly, the range of energy flux densities delivered by the heavy particles shown in Fig. 11 should be considered an upper limit.

Most V₂O₅ films deposited in this work were crystalline, with the exception of films deposited at elevated pressures with inductively coupled plasmas, which saw a significant decrease in crystalline character. This processing space also corresponds to the lowest measured energy flux densities. The results can be used to infer the energy flux densities necessary for the onset and saturation in the crystalline character of V₂O₅ during PEALD across all parameters. Those values are estimated to be ≈ 0.02 and ≈ 0.04 W/m², respectively (depicted as the lower and upper boundaries in Fig. 11). All the films deposited under conditions above the upper critical value showed strong Raman signatures of crystalline V₂O₅, those deposited between the critical values showed weak signs of crystalline character, and those below the lower critical value showed no signatures of crystalline material (represented as “Crystalline,” “Mixed Crystalline Character,” and “Amorphous” in Fig. 11, respectively). An exception to this claim is the data corresponding to the film

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FIG. 7. (a) Raman spectra of VO_x as a function of pressure during the plasma step for plasmas operated in an inductively coupled mode. (b) A comparison of Raman spectra of VO_x deposited with inductively and capacitively coupled plasmas at 5000 mTorr. XPS of (c) the O 1s and V 2p core levels as a function of pressure (plasmas operated in an inductively coupled mode unless otherwise noted) and (d) C and N contamination as a function of pressure during the plasma step.

produced at a plasma power of 200 W where the measured energy flux density was below the upper energy flux density threshold necessary for the saturation in the material's crystalline character. As previously mentioned, the coupling nature of the plasma (capacitive versus inductive coupling) varied over time. The film deposited at 200 W was done so with a capacitively coupled plasma, while the Langmuir probe data were acquired at a later date, when the plasma was operating in an inductively coupled mode, despite the same applied power. Understanding that capacitively coupled plasmas are capable of higher energy flux densities at relatively low power can explain our observance of crystalline V_2O_5 under these conditions.

Across the numerous datasets represented here, it is worth noting that the calculated energy flux densities have not considered other means of energy delivery such as photons, internally excited species (e.g., metastables) that release energy upon impact, and the release of energy associated with the neutralization of ions upon impact. It is also important to note that this value is dependent on the experimental conditions used during deposition, such as the temperature setpoints. Although temperature was held constant throughout this work, it is reasonable to assume that this threshold energy flux density for crystallization would decrease in a higher temperature environment. Overall, the combined results shown in Fig. 11 suggest that for all

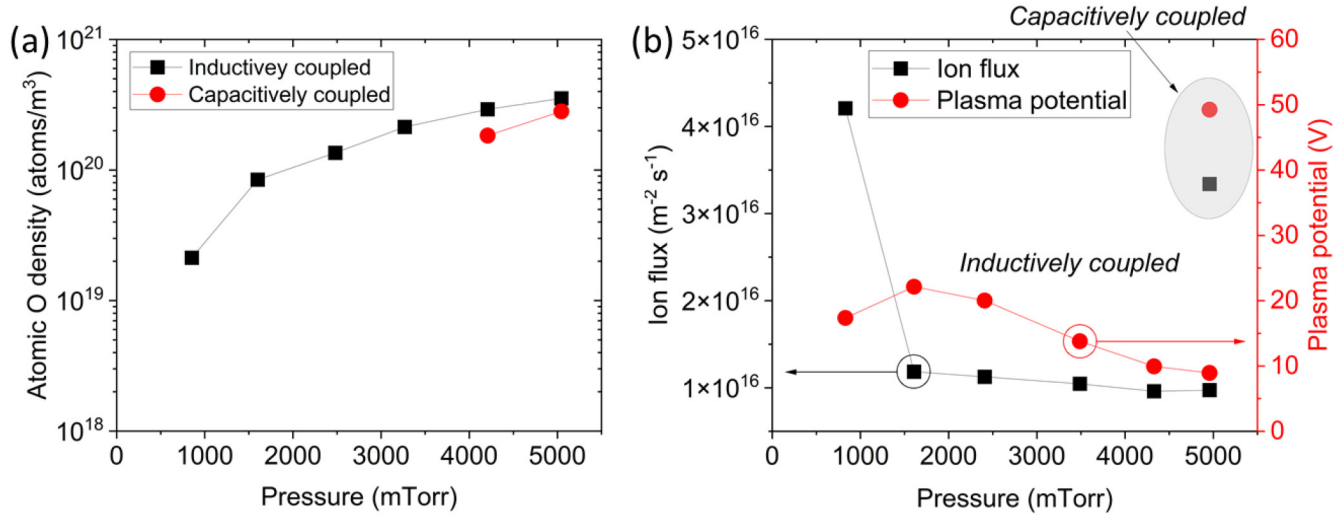


FIG. 8. (a) Atomic O density and (b) ion flux and plasma potential as a function of chamber pressure during plasma operation. Some conditions measured in both inductively and capacitively coupled modes.

variables explored, only an increase in pressure with the use of an inductively coupled plasma produced an energy flux density below the critical value necessary to fully develop the material's crystalline character, thus explaining why most films were crystalline.

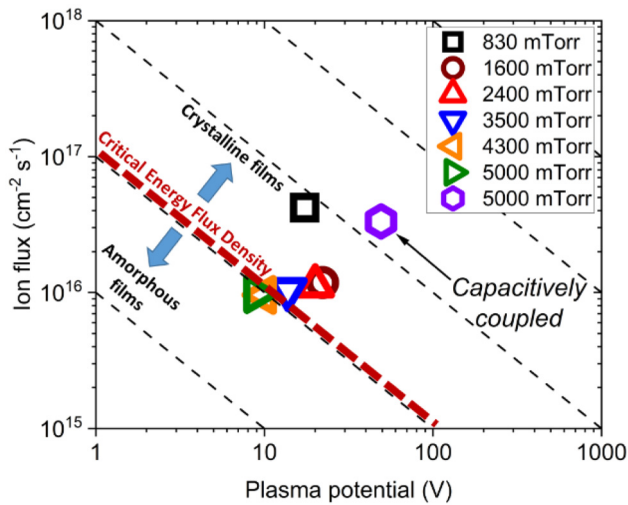


FIG. 9. Plot of the ion flux and plasma potential, following the arguments of Boris *et al.* (Ref. 2). The data represent the measured experimental data from this work for inductively coupled plasmas (unless otherwise noted). Thin dashed black lines represent the lines of constant energy flux density. The bold red dashed lines represents the approximated critical energy flux density necessary for the onset of crystallization.

The above discussion assumes the amorphous-to-crystalline transition is driven by the energy deposition rate, which implies a heating process. That is, the temperature at the surface of the material rises when the energy deposition rate exceeds the energy dissipation rate. When the temperature exceeds a threshold value, the material will crystallize. A recent review by Arts *et al.* presents an alternative critical parameter, where the driver of structural changes (e.g., crystallinity, sputtering, and dislocations) is the kinetic energy of incoming particles.²² Here, when the

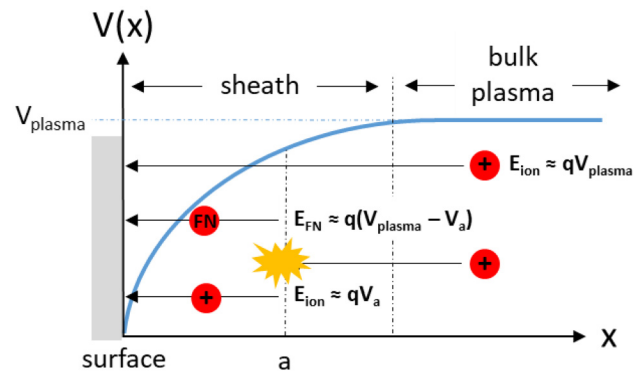


FIG. 10. Illustration depicting the effects of collisions on ion transmission across a sheath and energy deposition at the surface. For ions that cross the sheath without collisions, their kinetic energy when impacting the surface will be commensurate with the plasma potential. If an ion undergoes a charge exchange collision while crossing the sheath, a fast neutral (FN) and a slow ion will be created. The kinetic energy of each particle when hitting the surface will depend on the potential at the location of the charge exchange reaction ($x = a$).

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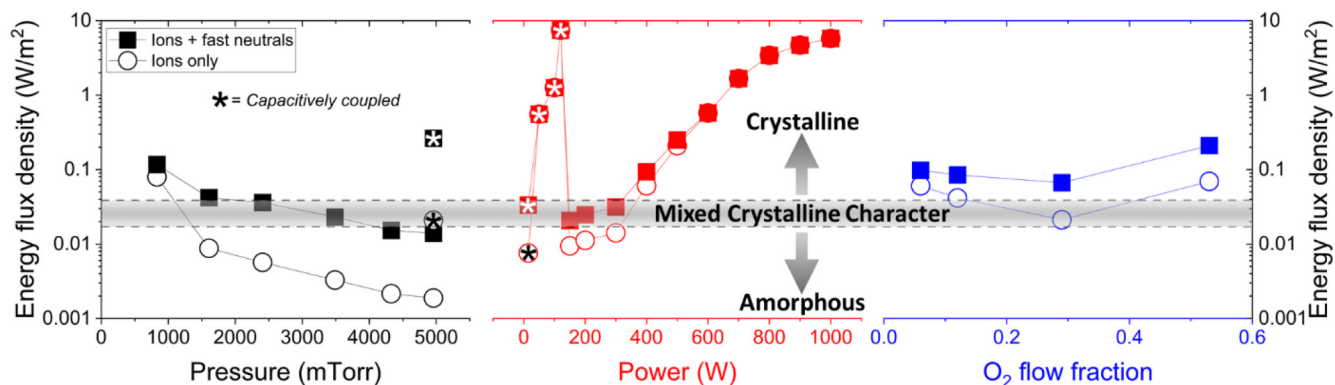


FIG. 11. Calculated energy flux density delivered to the surface via ions (open circles) as well as ions and fast neutrals (squares) plotted as a function of pressure, power, and O_2 flow fraction. The dotted lines represent the estimated energy flux density from ions and fast neutrals necessary for the onset of crystallization ($\approx 0.02 \text{ W/m}^2$) and saturation in crystalline character ($\approx 0.04 \text{ W/m}^2$). The area of *Mixed Crystalline Character* showed weak evidence of the crystalline material. The boundaries for crystallization thresholds are estimated from thin film characterizations.

incoming particle energy exceeds a threshold energy value, the material will crystallize locally. The difference being the particle flux or rate of delivery is not critical. Instead, the ion fluence (or dose)—the flux integrated over the exposure time—determines the extent of crystallization. To examine this, we can recast the results of Fig. 11 in terms of the calculated mean ion energy [Eq. (2)] and the measured plasma potential. The latter represents the maximum ion energy at a given condition and is likely the more significant parameter if crystallization is, indeed, limited by the kinetic energy of impinging ions. Those results are shown in Fig. 12, where the ordinate is the kinetic energy of ions that strike the surface. The results suggest a threshold energy for the onset of crystallization ranging between about 2 and 11 eV, depending on which energy is considered.

When comparing the results of Figs. 11 and 12, we see similar trends. That is not surprising since the scaling of the ordinate in the two figures differs by the ion flux. Unfortunately, the results provide little insights as to which metric is critical in driving crystallinity. When considering fully developed crystalline films produced across all parameters, both the energy flux density and ion energies are substantially larger than those that produced amorphous material at the highest pressure. For the films with mixed crystalline character produced in the mid-pressure range, it is more challenging. Here, the ion energy might be sufficiently high, but the ion fluence is too low to fully crystallize the film. Similarly, the energy flux density might be insufficient to fully saturate the crystalline character of the films on the timescales explored in this work. Additional studies are required to better understand the results.

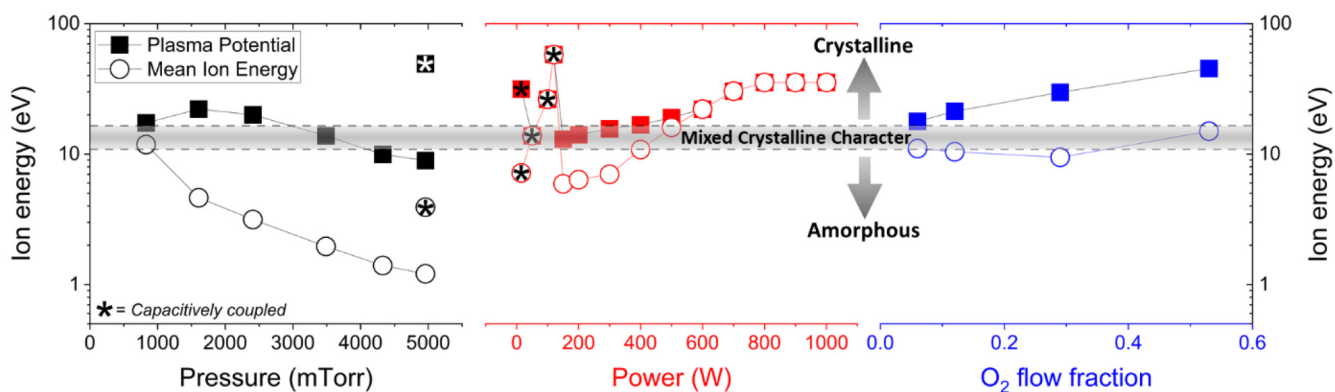


FIG. 12. Ion energies derived from the measured plasma potential (squares) and from the calculated average ion energy (circles) plotted as a function of pressure, power, and O_2 flow fraction. The dotted lines represent the estimated ion energy necessary for the onset of crystallization ($\approx 11 \text{ eV}$) and fully developed crystalline character ($\approx 17 \text{ eV}$). The area of *Mixed Crystalline Character* showed weak evidence of the crystalline material. The boundaries for crystallization thresholds are estimated from thin film characterizations.

IV. SUMMARY AND CONCLUSIONS

The work presented here investigates the plasma properties during the deposition of VO_x by PEALD. The effect of the plasma step duration, pressure, power, gas chemistry, and coupling mode is studied. In the majority of cases, crystalline V_2O_5 is produced. However, by controlling the pressure and plasma coupling mode in the system during the oxidizing plasma half-step, the crystallinity of the film can be controlled. It was found that inductively coupled plasmas operating at higher chamber pressures resulted in less crystalline films, and in some cases, this was despite being chemically indistinguishable from the crystalline films. This effect was found to be a consequence of a decrease in the ion flux and plasma potential with increasing pressure, which ultimately leads to a reduction in the ion energy and energy flux density delivered to the sample. In the extreme case of an inductively coupled plasma operated at 5000 mTorr, a minimum energy flux density and ion energy were calculated, no evidence of crystallinity was observed, and the films showed high levels of C and N, suggesting incomplete ligand exchange due to enhanced attenuation of the energy delivered by the remote plasma. In contrast, the effect of plasma power and gas chemistry had little effect on the crystallinity of the deposited thin films. It was concluded that this was due to the energy flux density and ion energy remaining above the threshold for crystallization.

Across the plasma parameters investigated in this work, no control over the V-O stoichiometry was achieved. This result is consistent with other works that demonstrate the formation of only V_2O_5 in the presence of oxidizing plasmas. This may be due to the relatively high concentrations of reactive atomic oxygen species produced in O_2/Ar plasmas. It can be inferred that a critical threshold of the atomic O density exists for the formation of V_2O_5 but is lower than all conditions tested in this work. In order to produce crystalline substoichiometric VO_x compounds, such as VO_2 , different approaches are needed. These may require a significant reduction in the partial pressure of oxygen during the plasma half-step, which in many cases may require hardware modifications to typical ALD reactors. Alternatively, an A-B-C ALD process incorporating H_2 plasma reductions may be useful since H_2 plasmas have previously been shown to reduce the surface of V_2O_5 films.²³

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

P. M. Litwin: Conceptualization (equal); Data curation (equal); Formal analysis (equal); Investigation (equal); Writing – original

draft (lead). **N. Nepal:** Writing – review & editing (equal). **M. G. Sales:** Writing – review & editing (equal). **D. R. Boris:** Formal analysis (equal); Investigation (equal); Writing – review & editing (equal). **M. J. Johnson:** Formal analysis (equal); Investigation (equal); Writing – review & editing (equal). **S. G. Walton:** Conceptualization (equal); Formal analysis (equal); Funding acquisition (equal); Writing – review & editing (equal). **V. D. Wheeler:** Conceptualization (equal); Formal analysis (equal); Funding acquisition (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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